



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2025-26

Course: Introduction to Web Technology

Course code: 2344211

Year: SE SEM: IV

Experiment No. 4
Aim : Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.
Name: Hetvi Patel
Roll Number: 56
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	20	

Checked by

Name of Faculty :



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Signature :

Date

Experiment 4

Title

Develop a dynamic webpage where content updates on button click using JavaScript DOM.

Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

Theory:

1. JavaScript and the DOM (Document Object Model):

- The **DOM** is a hierarchical representation of the elements on a webpage. It allows programming languages like JavaScript to interact with the structure, style, and content of the page.
- Using **DOM methods**, we can:
 - Access elements (getElementById, getElementsByClassName, querySelector)
 - Change content (innerHTML, textContent)
 - Modify attributes (setAttribute, removeAttribute)
 - Change styles dynamically (style.property)
- Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.

2. Dynamic Webpage Content:

- A **dynamic webpage** changes content without reloading the page.
- Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.



- In this lab, content updates are achieved using `document.getElementById().innerHTML` to manipulate text dynamically.

○

3. Introduction to jQuery:

- **jQuery** is a lightweight, fast JavaScript library that simplifies:
 - HTML document traversal and manipulation
 - Event handling
 - Animations and effects
- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax Example:

`$(selector).action();`

- `$()` is the jQuery function
- `selector` identifies the HTML element
- `action()` is the method applied (like `hide()`, `show()`, `animate()`)

4. jQuery Effects and Animations:

- **Hide/Show:** Instantly hide or show elements.
- **Toggle:** Switch visibility state of an element (hide → show, show → hide).
- **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
- **Benefits of jQuery Effects:**
 - Improves visual interactivity
 - Reduces manual CSS manipulation
 - Provides smoother user experience



5. Integration of JS DOM and jQuery:

- Using **pure JavaScript** allows control over element content and structure.
- Using **jQuery** complements JS by handling animations and effects more efficiently.
- Combining both allows for **dynamic and interactive web pages** where content changes and user interactions are smooth and visually appealing.

6. Applications:

- Dynamic webpages are used in:
 - Online forms that validate inputs instantly
 - Content sliders and carousels
 - Real-time notifications
 - Interactive dashboards

Procedure

1. Include jQuery CDN:

```
<script src="https://code.jquery.com/jquery-3.7.0.min.js"></script>
```

2. Create HTML element:

```
<div id="box">Hello</div>  
<button  
id="btn">Toggle</button>
```

3. Apply jQuery:

```
<script>  
$('#btn').click(function(){  
    $('#box').toggle(500);  
});  
</script>
```



Scenario-Based Exercises:(jQuery Lab Task)

Design a simple interactive College Notice Board webpage using HTML, CSS, JavaScript, and jQuery. The page should display a default notice message and include buttons that update the notice text dynamically using JavaScript DOM when clicked. Use jQuery to hide and show the notice, toggle its visibility with a single button, and animate the notice by moving it or changing its appearance (such as font size or opacity). The webpage should work without page refresh and demonstrate basic DOM manipulation and jQuery effects.

Code :

```
<!DOCTYPE html>

<html>

<head>

    <meta charset="UTF-8">

    <title>Dynamic Webpage</title>

    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

    <style>

        body {

            font-family: Arial, sans-serif;

            text-align: center;

            margin-top: 40px;

            background-color: #f5f5f5;

        }
```



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```
#box {  
    width: 150px;  
    height: 150px;  
    background-color: teal;  
    margin: 20px auto;  
    position: relative;  
    border-radius: 10px;  
}
```

```
button {  
    padding: 10px 15px;  
    margin: 5px;  
    cursor: pointer;  
    border: none;  
    background-color: teal;  
    color: white;  
    border-radius: 5px;  
}
```

```
button:hover {
```



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```
background-color: darkcyan;  
}
```

```
#content {  
    font-size: 18px;  
    margin: 20px;  
    color: darkblue;  
    font-weight: bold;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Dynamic Webpage</h2>
```

```
<div id="content">Click a button to interact with the page</div>
```

```
<button id="changeBtn">Change Content</button>
```

```
<button id="hideBtn">Hide</button>
```

```
<button id="showBtn">Show</button>
```



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```
<button id="toggleBtn">Toggle</button>
```

```
<button id="animateBtn">Animate</button>
```

```
<div id="box"></div>
```

```
<script>
```

```
$(document).ready(function(){
```

```
    $("#changeBtn").click(function(){
```

```
        $("#content").text("Content updated using JavaScript DOM!");
```

```
    });
```

```
    $("#hideBtn").click(function(){
```

```
        $("#box").hide(800);
```

```
    });
```

```
    $("#showBtn").click(function(){
```

```
        $("#box").show(800);
```

```
    });
```

```
    $("#toggleBtn").click(function(){
```




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```
$("#box").toggle(800);

});

$("#animateBtn").click(function(){

    $("#box").animate({

        left: "+=200px",

        opacity: 0.5,

        height: "200px",

        width: "200px"

    }, 800).animate({

        left: "0px",

        opacity: 1,

        height: "150px",

        width: "150px"

    }, 800);

});

});

</script>

</body>
```

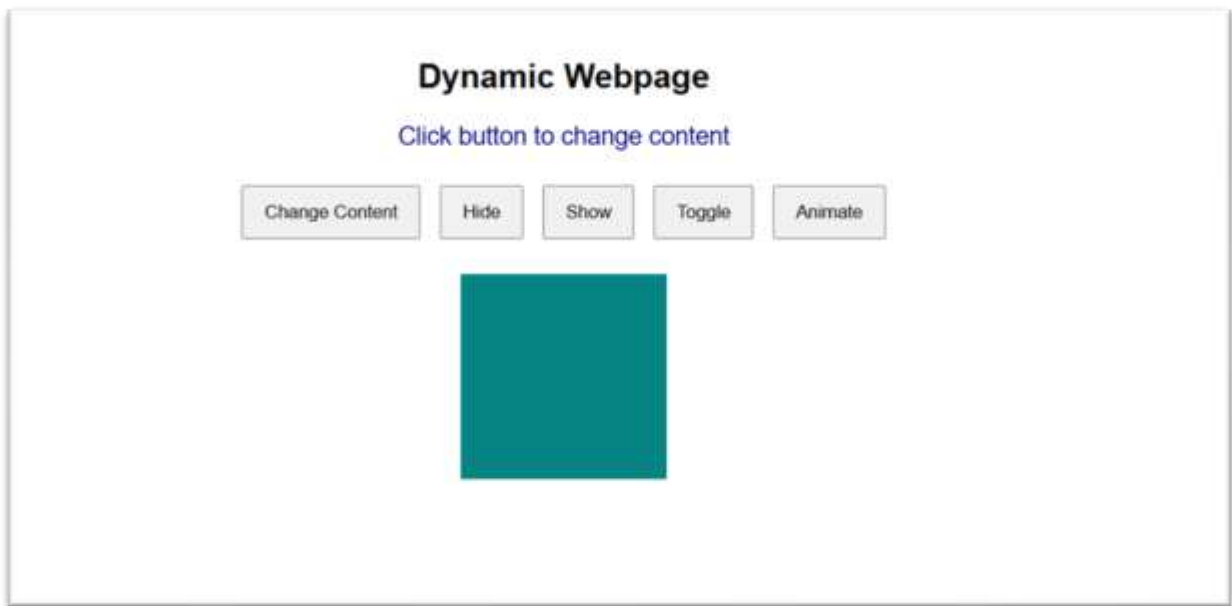


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</html>

Output:



Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation of content updates, visibility toggling, and animations.
